ECG Signal Preprocessing for Smart Wearables: Filtering Pipeline for Reliable HR/HRV Analysis

Abstract—**Music has been shown to have strong effects on the human brain, body, and emotions. People experience measurable physical responses while listening to music. Previous studies indicate that brain activity, emotional states, and heart functions can change with music exposure. This highlights the connection between sounds and physical reactions [1]-[4]. Heart rate and heart rate variability (HRV) are important indicators for tracking emotional and stress responses to music [5]. It is crucial to properly acquire and process ECG signals for effective emotion recognition and health applications. Techniques in signal processing, like higher-order Butterworth filters, notch filtering, and adaptive artifact removal, help improve ECG quality for better analysis of physiological markers [6]. At the same time, deep learning has transformed music signal processing, supporting applications like music information retrieval, emotion recognition, and automatic music creation [7].**

This work explores how listening to music affects emotional changes through heart rate signals. We apply advanced ECG preprocessing methods and analyze HRV in response to music to discover patterns that show emotional changes. We connect knowledge from biomedical signal processing and music deep learning to better understand how music impacts both physiology and emotions.

Keywords—**Music and Emotion Recognition, Heart Rate Variability (HRV), Electrocardiogram (ECG) Signal Processing, Butterworth Bandpass Filter, Notch and Adaptive Filtering, Physiological Signal Analysis, Music-Induced Mood Changes, Deep Learning in Music Signal Processing, Biomedical Signal Preprocessing, Emotion Detection Systems**

I. Introduction

Music has long been seen as a universal experience that crosses cultural and language boundaries. It deeply affects human thought, body functions, and emotions. Studies have consistently shown that music can trigger strong emotional reactions by influencing brain activity, memory, and attention [1]. Research into the brain has shown that our physiological and psychological reactions to music can be grouped into clear emotional categories. This underscores music's ability to impact our moods [2].

Among the various physical signs, heart-related responses have proven to be reliable markers for studying emotional changes caused by music. Heart rate (HR) and heart rate variability (HRV) are sensitive signs of emotional arousal, stress management, and the balance of the autonomic nervous system [3]. Studies indicate that musical features like tempo, rhythm, and harmony directly affect the body's parasympathetic and sympathetic systems, changing cardiovascular patterns in noticeable ways [4]. As a result, ECG-based HRV analysis has become a trusted method in recognizing emotions, especially when examining feelings caused by music.

Several experimental studies have confirmed the direct impact of music on heart responses. For example, research has found that listening to different music genres can significantly change HRV features, supporting the idea that strong links exist between physical and emotional states [5]. These insights have led to combining heart monitoring with affective computing, which quantifies emotions using both subjective feedback and objective bodily signs.

However, effective HRV analysis relies heavily on the quality of ECG recordings. Raw ECG signals are often distorted by factors like baseline shifts, movement interference, muscle noise, and power line issues. These distortions can hide important information. Therefore, preprocessing is crucial to ensure that clear and accurate heart rate data is extracted. Saputri [6] showed that an 8th-order Butterworth Infinite Impulse Response (IIR) filter could deliver better clarity of ECG signals compared to traditional methods, especially in reducing baseline shifts and high-frequency noise. These preprocessing advancements are essential for guaranteeing trustworthy HRV and emotional analysis.

At the same time, significant progress has occurred in music signal processing through the use of deep learning techniques. Methods in music information retrieval, emotion detection, and automatic classification have improved with convolutional and recurrent neural networks, allowing for better feature extraction and understanding [7]. These developments create opportunities for interdisciplinary collaboration, where audio features from music can be compared with physical responses, helping us understand how outside stimuli influence our internal states.

In this study, we propose a method for exploring emotional changes caused by music using ECG-based HRV analysis. We subject raw ECG signals to a detailed preprocessing pipeline that includes an 8th-order Butterworth bandpass filter (0.5–40 Hz, SOS form), a 50 Hz notch filter (2nd order, SOS form), and an adaptive filter to reduce motion-related interference. Figure X illustrates how the raw noisy ECG signal is gradually refined to create a clean signal suitable for HRV extraction. We then analyze the processed ECG signals for HRV features and correlate them with emotional changes seen during controlled music exposure sessions. By merging advanced biomedical signal processing with insights from music analysis driven by deep learning, this study aims to enhance emotion recognition frameworks, where music acts not only as a stimulus but also as a possible therapeutic tool.

Classical $\rightarrow$ HR goes down Rock $\rightarrow$ HR goes up Favorite song $\rightarrow$ HR goes up Tempo HR driver $\rightarrow$ Emotional content is more important. Measure heart rate (BPM) changes before/after clips. Track HR variability (HRV) to get deeper insight into autonomic nervous system shifts. Compare across music types to see patterns (relaxing $\rightarrow$ HR$\downarrow$, stimulating $\rightarrow$ HR$\uparrow$).

II. Related Studies

Several studies have shown that music impacts brain activity, feelings, and heart responses [1–5]. Heart rate variability (HRV) from ECG is often used to measure emotional changes, but the quality of the signal is important for accurate analysis [6]. Deep learning has improved music signal processing and emotion detection [7]. However, most studies treat music and physiological signals separately and only use basic ECG preprocessing. Few studies combine deep learning with HRV analysis for emotion recognition. This study merges better ECG processing, HRV analysis, and music deep learning to gain a clearer understanding of emotional responses.

A. Multimodal Emotion Detection

Multimodal frameworks have been widely studied for music emotion recognition (MER). One study combined acoustic features of music, such as rhythm and timbre, with physiological measures that included heart rate, pulse of blood volume, skin conductance and skin temperature. This approach improved emotion classification, although some information was lost due to feature reduction [8]. In another study, researchers used signals such as ECG, GSR, and pulse detection to connect music moods with human emotions. This research showed that the combination of multimodal signals improves the accuracy of emotion recognition, although issues such as poor signal quality and challenges with wearable technology were noted [9]. Overall, these studies emphasize the potential of multimodal frameworks while also acknowledging the need for better preprocessing and improved hardware.

B. EEG-Based Music Emotion Studies

A significant number of studies have focused on electroencephalography (EEG) to detect emotions induced by music. For instance, one study looked into the neural entrainment and spatial modulation hypotheses, finding that EEG alpha-band power (8–12 Hz) has a strong correlation with selective attention to melodies. By extracting features with Common Spatial Patterns (CSP) and using Support Vector Machine (SVM) classifiers, this study accurately recognized attended streams, although the small dataset limited its broader applicability [10]. Another study extracted spectral features from EEG signals and used artificial neural networks for emotion classification. This approach proved to be robust against artifacts due to bandpass and notch filtering [11].

In addition to experimental research, reviews have highlighted the relationship between EEG bands (alpha, beta, gamma) and emotional states like arousal and relaxation. While these reviews provide useful theoretical insights, they often lack detailed descriptions of signal-processing methods [12]. Moreover, research on how musical tempo affects EEG connectivity revealed that slow, medium, and fast tempos impact valence and arousal dimensions differently [13]. Recently, EEG signals have been integrated into recommender systems to predict personal music preferences. Results showed improved prediction accuracy but also pointed out issues like reduced signal quality from portable dry-electrode devices [14]. These studies highlight the strengths of EEG in MER and also point to limitations regarding dataset size, signal quality, and generalizability.

C. ECG and Heart-Signal Processing in Music Studies

Beyond EEG, researchers have extensively studied music's effect on cardiovascular signals. One study applied an 8th-order Butterworth IIR filter to ECG signals. This improved the clarity of PQRST waveforms and allowed for more accurate heart rate variability (HRV) analysis, though some baseline drift was still present [15]. Another study looked at how different music genres affect heart rate. It found that tempo was not the main factor influencing heart rate variation. Instead, emotional content and genre were more significant, with classical music reducing heart rate and rock music increasing it [16].

A broader survey of ECG analysis methods emphasized that preprocessing (typically through bandpass filtering), HRV-based feature extraction, and classification are central to ECG studies on music's effects. However, the authors noted that traditional processing methods face challenges compared to machine learning-based techniques [17]. Expanding beyond just heart rate, one study combined ECG and EEG to investigate heart–brain coupling through complexity measures like fractal dimension and sample entropy. While this revealed meaningful correlations, the study acknowledged that nonlinear dynamics are not fully captured with these methods [18]. Together, these works present ECG as a reliable physiological tool for emotion-related music studies, especially when paired with effective signal processing or multimodal frameworks.

D. Deep Learning and Computational Frameworks

Deep learning has become more common in music emotion recognition and analyzing physiological signals. Reviews have discussed applications of architectures like CNNs, RNNs, LSTMs, autoencoders, and GANs for tasks ranging from music classification to emotion detection. These methods show better feature learning compared to traditional processes, but they still struggle with issues of limited datasets and lack of interpretability [19]. To overcome some of these obstacles, algorithm unrolling has been proposed as a structured framework that links model-based iterative algorithms with neural networks. This approach offers both interpretability and efficiency, but it sacrifices some flexibility [20].

Simultaneously, deep learning has been applied to physiological signal processing. For example, a framework called DeepHeart used CNNs to denoise photoplethysmogram (PPG) signals and applied spectral calibration for heart rate estimation. This method achieved high accuracy but required significant computational resources, limiting its use in real-time wearable devices [21]. These studies demonstrate that while deep learning holds potential for advancing MER, practical implementation needs to address data availability and computational efficiency issues.

E. Correlated Brain–Skin Responses to Music

Some studies have used skin-based responses in music emotion research in addition to EEG and ECG. For example, one study combined EEG and galvanic skin response (GSR) while participants listened to various genres like relaxing music, pop, and rock. By using fractal dimension and entropy analysis, the research identified physiological responses specific to each genre. However, its general applicability was limited due to a small sample size and narrow range of music genres [22]. These findings highlight the potential for including peripheral physiological signals in multimodal MER systems.

F. Standardization and Broader Perspectives in MER

Lastly, broader reviews have stressed the need to standardize frameworks for music emotion recognition. One position paper suggested that emotion models like the valence–arousal space should form the basis for MER research and that personal context and listener variability should be included in future systems. The study also emphasized the importance of validating across datasets to ensure broad applicability. However, since the paper did not have empirical validation, its recommendations remain more theoretical than experimental [23].

G. Research Gap

Although there has been progress in understanding how music affects brain activity and cardiovascular responses, some limitations still exist in current studies.

Limited Integration of Music and Cardiovascular Signals The influence of music on brain function [1,2] and its effects on cardiovascular markers such as HRV [3–5] have often been studied separately. A systematic approach combining music exposure with ECG-derived HRV analysis for emotion detection has not been widely applied.

Challenges in ECG Signal Quality ECG signals frequently suffer from baseline drift, noise, and motion artifacts, which undermine the reliability of HRV-based emotion recognition. While the use of an 8th-order Butterworth filter has been suggested [6], many studies continue to rely on simpler filtering methods, limiting the accuracy of feature extraction.

Underutilization of Advanced Signal Processing Pipelines Many studies have used standard bandpass and notch filters; however, comprehensive multi-stage pipelines that incorporate adaptive filtering are rare. Consequently, leftover artifacts can compromise the interpretation of HRV.

Insufficient Use of Deep Learning for Music–Emotion Fusion Although deep learning has improved music information retrieval and emotion detection from audio features [7], it has not been fully integrated with ECG-derived HRV features. This gap has limited the development of interdisciplinary approaches that connect music perception with physiological responses in real time.

This project tackles several of the previously mentioned limitations. It introduces a structured pipeline for ECG preprocessing, incorporating an 8th-order Butterworth bandpass filter, a 50 Hz notch filter, and an adaptive filter. This method effectively reduces baseline drift, electrical interference, and motion artifacts, ensuring quality ECG signals for accurate HRV extraction. Unlike previous research, this study directly ties music exposure with ECG-derived HRV analysis, enabling an objective investigation of emotional changes through physiological signals. Furthermore, it connects advancements in music deep learning, fostering the integration of biomedical signal processing with computational music analysis. Through these contributions, this work lays a stronger foundation for emotion recognition, using music both as a stimulus and a possible therapeutic tool.

III. Methodology

A comprehensive ECG signal processing pipeline is implemented in this project to simulate, filter, and clean heart signals under varying physiological conditions. Synthetic ECG signals for different heart rates, noise levels, and arrhythmia-like scenarios are generated using NeuroKit2. The preprocessing pipeline consists of three main stages: first, the meaningful ECG frequency band (0.5–40 Hz) is isolated using an 8th-order Butterworth bandpass filter, thereby removing baseline wander and high-frequency noise. Next, powerline interference at 50 Hz is removed using a manual 2nd-order notch filter, ensuring the elimination of periodic artifacts. Finally, an adaptive Least Mean Squares (LMS) filter is applied, in which weights are dynamically adjusted based on the error between the noisy input and filtered output to further suppress residual noise and motion artifacts. Through the combination of these filters, a clean ECG signal is produced, which is suitable for downstream analysis such as heart rate variability (HRV) computation or correlation with music-induced physiological responses. Visualization of each stage allows the assessment of filter performance and the effectiveness of the adaptive LMS in minimizing the mean squared error of the ECG signals.

A. Dataset Generation

Simulated ECG signals were created using the NeuroKit2 library.Noise and baseline motion artifacts were artificially added to resemble realistic ECG recordings. Heart-rate patterns were generated to match emotional responses:

MUSIC TYPE	MEAN HEART RATE	EXPECTED STRESS
CLASSICAL	62±3	LOW
FAVORITE	82±4	MEDIUM
ROCK	105±5	HIGH

Fig. 1: Music type

B. Filtering Mechanisms

In signal processing, filters are used to selectively allow certain frequencies to pass while unwanted frequencies are attenuated. Noise is removed, important frequency components are isolated, and signal features are enhanced for analysis. High-frequency noise is removed by low-pass filters, slow drifts are eliminated by high-pass filters, a specific frequency range is retained by band-pass filters, and a single interfering frequency is removed by notch filters. In ECG processing, the main heart signal (0.5–40 Hz) is retained by a band-pass filter, 50 Hz powerline interference is eliminated by a notch filter, and residual noise is further reduced by an adaptive LMS filter, resulting in a clean signal suitable for analysis.

1) Bandpass Filtering: To remove baseline wander and high-frequency noise, a Butterworth bandpass filter of order 8 was applied in the range 0.5–40 Hz. General digital Butterworth filter transfer function:

$$H(s) = \frac{1}{1 + \left(\frac{s}{\omega_c}\right)^{2N}}$$

where:

$$N = \text{filter order (here, 4)},$$
$$\omega_c = \text{cutoff frequency (normalized)}.$$

2) Notch Filtering: To suppress 50 Hz powerline interference, a digital IIR notch filter was designed. Transfer function:

$$H(z) = \frac{1 - 2\cos\left(\frac{2\pi f_0}{f_s}\right) z^{-1} + z^{-2}}{1 - 2r\cos\left(\frac{2\pi f_0}{f_s}\right) z^{-1} + r^2 z^{-2}}$$

$$r = 1 - \frac{1}{2Q}$$

where:

$$f_0 = 50 \text{ Hz} \quad \text{(notch frequency)}$$

$$f_s = 250 \text{ Hz} \quad \text{(sampling frequency)}$$

$$r = 1 - \frac{1}{2Q}, \quad Q = 30 \text{ (quality factor)}$$

This removes a narrow band around 50 Hz without affecting neighboring frequencies.

3) Adaptive Baseline Filtering: An adaptive baseline filter is designed to remove baseline drift or low-frequency noise from signals such as ECG by dynamically adjusting its parameters in response to changing signal characteristics.

Adaptive filter output:

$$b[n] = \frac{1}{W} \sum_{k=0}^{W-1} x[n-k]$$

$$y[n] = x[n] - b[n]$$

where:

$$x[n] = \text{raw ECG signal}$$

$$b[n] = \text{estimated baseline (moving average)}$$

$$y[n] = \text{baseline-corrected ECG}$$

$$W = \text{window size (here, } W = 300 \text{ samples)}$$

C. R-Peak Detection

R-peaks were detected using NeuroKit2's ECG peak detector, and RR intervals were calculated as:

$$RR_i = \frac{R_{i+1} - R_i}{f_s}$$

where:

$$R_i = \text{index of } i\text{-th R-peak},$$
$$f_s = \text{sampling frequency}.$$

D. Time-Domain HRV Features

Time domain HRV (Heart Rate Variability) filters focus on analyzing the variations in time intervals between successive heartbeats (RR intervals) directly. Time domain filtering helps remove noise and baseline drift from ECG signals, enabling the precise calculation of these temporal HRV features. Such filters are preferred for short-term recordings and provide essential insights into autonomic nervous system function by quantifying beat-to-beat variability.

1) Mean NN Interval::

$$\text{MeanNN} = \frac{1}{N} \sum_{i=1}^{N} RR_i$$

2) Standard Deviation of NN Intervals (SDNN)::

$$\text{SDNN} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (RR_i - \text{MeanNN})^2}$$

3) Root Mean Square of Successive Differences (RMSSD)::

$$\text{RMSSD} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (RR_{i+1} - RR_i)^2}$$

4) pNN50::

$$\text{pNN50} = \frac{\#(|RR_{i+1} - RR_i| > 50 \text{ ms})}{N}$$

5) Sample Entropy: where:

$$A = \text{matches of template length } (m+1),$$
$$B = \text{matches of template length } m.$$

E. Raw ECG Statistical Features

Mean:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Standard Deviation:

$$\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu)^2}$$

1) Data Standardization:

$$z = \frac{x - \mu}{\sigma}$$

F. Machnie learning Modals

A file that represents a system that has been trained to recognize patterns in data

1) Voting Classifier (Music Emotion Detection):

$$\hat{y} = \arg\max_{y}\left(\sum_{k=1}^{K} w_k \cdot P_k(y \mid x)\right)$$

where:

$$P_k(y|x) = \text{probability from classifier } k,$$
$$w_k = \text{model weight.}$$

2) Voting Regressor (Stress Prediction):

$$\hat{S} = \frac{1}{K}\sum_{k=1}^{K} f_k(x)$$

where:

$$f_k(x) = \text{stress prediction from regressor } k.$$

G. Mean Squared Error Minimization

To ensure accurate stress estimation, the regression model is trained by minimizing the mean squared prediction error. This guarantees that the predicted stress values remain as close as possible to the true stress values over time.

The cost function for the model is defined as:

$$J(w) = \mathbb{E}\{e^2[n]\},$$

where

$$e[n] = y[n] - \hat{y}[n]$$

is the prediction error, $y[n]$ is the actual stress value, $\hat{y}[n]$ is the predicted stress value, and $\mathbb{E}\{\cdot\}$ denotes the expectation operator.

The objective function ensures continuous minimization of the squared error between the actual and estimated stress levels.

The optimization process is governed by:

$$w(n+1) = w(n) - \mu\,\frac{\partial J(w)}{\partial w},$$

where μ represents the learning-rate parameter that regulates convergence speed and stability. The loss function is defined in terms of the expected squared error. Minimization of this cost is performed to improve prediction accuracy. A gradient-based update rule is employed, and the step size is controlled by μ.

H. Evaluation Metrics

Classification Accuracy

Classification accuracy was computed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives and false negatives respectively.

I. Regression Metrics

Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of the true target values.

J. Confusion Matrix

Confusion matrices were used for both music emotion classification and stress-level prediction.

References